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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2022-2023

Program: B Tech/MBA Tech | Year: II/III Semester: III/V

Stream : B. Tech (IT, Computer, CSE-Cyber Security, AI, AIML, AIDS), MBA Tech (IT, Computer, AI) , EXTC

Subject: Discrete Mathematics | **Time:** 3 hrs (11 am to 2 pm)

Date: 13/06/2023 | No. of Pages: 4

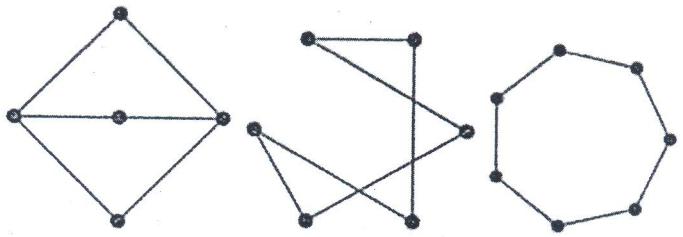
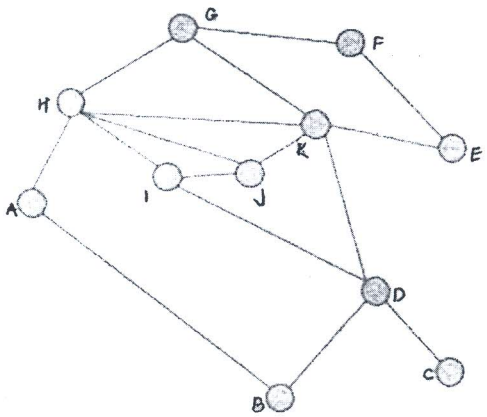
Marks: 100

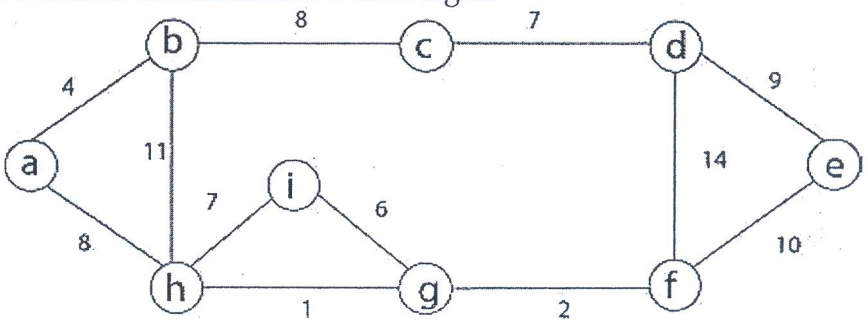
Special Re-Examination

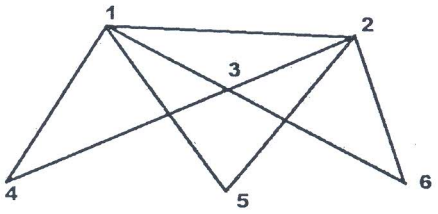
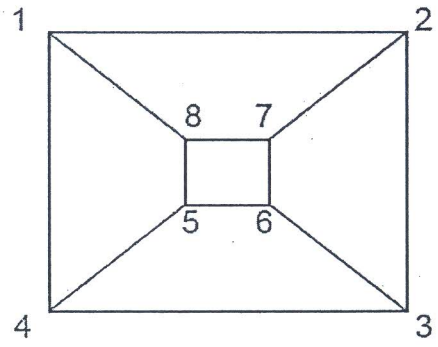
Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1. Question 1 is compulsory.
2. Solve any four out of remaining six questions
3. **In all, 5 questions to be attempted**
4. All questions carry equal marks.
5. **Answer to each new question to be started on a fresh page.**
6. **Figures in brackets on the right hand side indicate full marks.**
7. **Assume Suitable data if necessary.**
8. **Use of scientific calculator is allowed.**

		Answer the following:	Marks
CO1/SO2/BL3	Q.1 a.	If $U=\{1,2,3,4,5,6,7,8,9\}$, $A=\{1,2,4,6,8\}$, $B=\{2,4,5,9\}$ Find $A \cup B$, $A \cap B$, A^C and B^C.	[4]
CO2/SO1/BL2	b.	Construct a truth table for $p \rightarrow (q \wedge p)$.	[4]
CO1/SO1/BL1	c.	Prove that if 8 people are assembled in a room then 2 of them are born on the same day of a week.	[4]
CO4/SO1/BL3	d.	Check whether the following graphs exists. Justify your answer. i) A tree with 4 vertices and 3 edges. ii) A graph with degree sequence (2,2,3,3,5)	[4]

CO4/SO2/BL5	e.	Prepare composition table for G={1, -1, i, -i} where $i = \sqrt{-1}$. Also find identity if it exists.	[4]
CO3/SO2/BL5	Q.2 a.	Determine the sequence whose recurrence relation is $a_n = 2a_{n-1} - a_{n-2}$, $n \geq 3$, with initial conditions $a_1 = 1.5$, $a_2 = 3$	[6]
CO1/SO2/BL3	b.	Which of the graphs below are bi-partite? Explain your answers. (i) (ii) (iii) 	[6]
CO3/SO2/BL4	c.	Show that the relation R on the set $A = \{x \in Z / 0 \leq x \leq 12\}$, given by $R = \{(a,b) : a = b\}$ is an equivalence relation. Find the set of all elements related to 1.	[8]
CO2/SO1/BL2	Q.3 a.	Using the principle of mathematical induction show that for all positive integers $n \geq 1$, $3^n - 1$ is multiple of 2.	[6]
CO2/SO2/BL6	b.	Estimate the chromatic number of the following graph using Welch-Powell Algorithm 	[6]

CO1/SO1/BL4	c.	Check whether the function $f:\mathbb{Z}\to\mathbb{Z}$ *given by* $f(x)=x^3$ is i) injective ii) surjective iii) bijective. Justify your answers.	[8]
CO3/SO2/BL4	Q.4 a.	Prove that the set of rational numbers $(\mathbb{Q},\oplus,\otimes)$ is a ring under the operations defined by $a\oplus b=a+b+5$ & $a\otimes b=a+b+\frac{ab}{5}$	[6]
CO2/SO2/BL2	b.	Convert $p\vee\{\sim p\to(q\vee(q\to\sim r))\}$ into Disjunctive Normal Form.	[6]
CO3/SO1/BL5	c.	Solve the recurrence relation : $a_n - 8a_{n-1} = 5 + 14n$	[8]
CO2/SO1/BL5	Q.5 a.	If G is a group of real numbers under addition and G' is the group of positive real numbers under multiplication, show that the mapping defined by $f(x)=e^x$ is an isomorphism.	[6]
CO2/SO2/BL4	b.	Prove that $\sqrt{5}$ is an irrational number using method of contradiction.	[6]
CO3/SO1/BL3	c.	Construct a minimum spanning tree using Kruskal's Algorithm. Also find the minimum total weight. 	[8]
CO1/SO2/BL4	Q.6 a.	Check whether the following relation R is reflexive, irreflexive, symmetric, asymmetric and anti-symmetric. A={1,2,3,4}, R={(1,2),(1,3),(1,4),(2,3),(2,4),(3,4)}	[6]

CO4/SO1/BL3	b.	Identify whether the following graph is planar or not. Give reason in detail.  	[6]
CO4/SO1/BL6	c.	Let G = {1, 2, 3, 4, 5, 6, 7} i.e. $Z_8 - \{0\}$ (i) Prepare composition table with respect to '$\times_8$' (ii) Is it a group.	[8]
CO2/SO2/BL3	Q.7 a.	Apply laws of logic to establish the equivalence: $[(p \vee q) \wedge (p \vee \sim q)] \vee q \leftrightarrow p \vee q$	[6]
CO3/SO7/BL2	b.	Solve the following recurrence relation : $a_n = 5a_{n-1} - 6a_{n-2} + 7^n$	[6]
CO4/SO1/BL3	c.	Determine whether Eulerian circuit, Hamiltonian path and Hamiltonian cycle exists for the following graph. If so, find them.  	[8]